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Factors that drive market share and the oligopolistic character of cross-border B2C e-commerce: an agent-based scenario-analysis approach

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Abstract

Business-to-Consumer (B2C) e-commerce has become a dominant, continuously evolving force in global retail. Most consumers today make cross-border purchases in marketplaces, while enterprises have found great opportunities, and competition has become increasingly fierce. To understand B2C properties is an important endeavor. This paper achieves two goals: it develops a method to prove tendencies in a simulation and shows the oligopolistic nature of cross-border B2C e-commerce marketplaces. These achievements enhance the understanding of cross-border B2C e-commerce, by employing novel approaches: Social simulation, agent-based modeling (ABM) tools, theorem-proving techniques, and scenario analysis. The proving method began with an experimental design to explore the model's dynamics for parameters defined for specific scenarios of interest, and the agents' options were randomly selected for a significant number of runs (Monte Carlo experiment). This procedure allowed us to identify and prove the necessity of a tendency (the oligopolistic character of cross-border B2C e-commerce) and determine the factors that drive it. In the second part of the proof procedure, the persistence, independence of the agents' choices, and scaling validity of the tendency were shown by significantly increasing the number of random experiments and the number of simulated agents. The model was also validated. The developed method satisfactorily addresses some challenges of theorem-proving. In all these experiments, the variable of interest was market share. The resulting order of influence of the factors driving market share was recognition, product and service attributes of the marketplaces, and word of mouth. Surprisingly, word of mouth was the least important factor.

Keywords

Cross-border B2C e-commerce, oligopoly, market share, ABM, scenario analysis, practical proofs

1. Introduction

Cross-border Business-to-Consumer (B2C) e-commerce refers to the process by which firms sell goods and services to customers outside their home markets using information and communication technologies.[1] The cross-border B2C ecommerce market was valued at $1220.5 billion in 2023 and is expected to reach $8071.1 billion by 2032.[2] The continuous development of B2C cross-border e-commerce has led to more development opportunities for enterprises, and competition has become increasingly fierce.[3]

Most consumers today make cross-border purchases in marketplaces.[4] Cross-border B2C e-commerce in marketplaces presents several factors of complexity: e.g., the owner of the marketplace is not the seller of the products, and there are many individual sellers, whose behavior cannot be entirely controlled by the B2C marketplace.[3] The difficulties in controlling these factors make the decisions and activities of the B2C marketplaces uncertain. These factors necessitate the careful examination of B2C cross-border e-commerce using diverse approaches, including social simulation and agent-based modeling (ABM). In a previous study, we examined the ABM social simulation of Chinese cross-border B2C e-commerce in Spain for

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Chinese dominance of the cross-border B2C e-commerce worldwide by marketplaces like AliExpress.[4] This constitutes the first part of our research.[5] The model used in this study was qualitatively validated, demonstrating e-commerce's dominance in several marketplaces (AliExpress and Wish). AliExpress in particular dominates the virtual marketplace across a fairly broad range of parameters and simulations. Factors driving this dominance and market share were identified in our research. These findings motivated the current research, as it was often observed that those individuals, agents, or firms that start as large entities in a simulation tend to remain so throughout their duration. This outcome seems to be characteristic of the broader economy currently (e.g. mergers, acquisitions). This is probably a function of the importance of economies of scale and due to the scope of networked businesses[6] operating in the physical and virtual worlds. The market leader (i.e. the company with the highest market share) has several advantages over its competitors, such as brand-name recognition, economies of scale, and greater bargaining power with suppliers and channels of distribution, all of which make it easier to maintain its position.[7]

Other companies exist online, however, besides those based in China. US-based Amazon is the website with the largest cross-border B2C e-commerce volume worldwide.[4] Therefore, we want to focus more broadly on cross-border B2C e-commerce by including Amazon to verify whether market dominance by a few companies still exists.

The focus of the above-named previous paper was primarily to show the persistence of the tendency found in Chinese marketplaces. This paper extends the analysis to show the oligopolistic nature of marketplaces worldwide, including Amazon, by further developing the procedure used in the first paper. While the previous work focused on finding and proving a tendency, the new procedure also includes demonstrating the robustness of the conclusion under a broader and appropriately defined exploration of simulation trajectories (e.g. varying the number of replications and scaling the number of agents in the simulation).

In line with this, this paper has two goals: (1) to develop a method to prove tendencies in a simulation and (2) to demonstrate the oligopolistic nature of cross-border B2C e-commerce marketplaces. The method developed appropriately deals with several challenges of theorem-proving[8,9] by, e.g., effectively managing the complexity of the proof through constrained exploration of the simulation trajectories to the parameter subsets suggested by domain experts and statistically valid random agent choices. In addition, the method is expressive, flexible, scalable, easily integrated with other simulation tools, simple (as it does not require extensive formal methods expertise), and applicable to actual cases and evolving systems. The main limitation of our approach is that most exploration of the simulation trajectories is random (the Monte Carlo method) and domain-expert-knowledge oriented, resulting in a less-formal proof than most usual theorem provers found in the literature.

Along with an overview of related studies,[5] this paper presents additional information on the rationality and novelty of using ABM and social simulation. These simulations' strengths in social studies are also strongly supported by Squazzoni's[10] work. Other authors have described cross-border B2C e-commerce more generally. ABM and social simulation allow for the exploration of scenarios of the possible future configuration of a system and its environment. In particular, explicit representations of the interacting agents, ABM, and agent reasoning are helpful for the virtual study of interaction and decision-making in marketplaces. ABM and scenario analysis enable us to study the dynamics of decision-making and the possible "future words" of marketplaces. Our approach explicitly focuses on agents' reasoning and is based on surveys.

One approach to decision-making (especially agent reasoning) in the social simulation field is explained by Cohen's[11] endorsements theory; this approach has been widely used by Moss[12] and others, including Bruce Edmonds and Ruth Mayer, at the Center for Policy Modeling in Manchester, UK,[13] in diverse research projects and modeling situations. Nonetheless, the available simulation languages currently do not offer adequate facilities to represent the agents' decision-making process realistically. Therefore, we developed a model in Java (a general-purpose computer language)[14] to represent the B2C e-commerce environment, which includes a realistic consideration of the buyers' decision-making mechanism based on the endorsements scheme and supported by surveys. This model is available in the public repository GitHub.[14] Additional information about why we developed our own simulation tool rather than using existing simulation languages, such as NetLogo or Repast, can be found in our previous work.[5] One element of the survey is word of mouth (WOM), which allow buyers to share and recommend marketplaces. We were unable to find previous agent-based models that employed endorsements and WOM to represent these kinds of markets.

This study is organized as follows: the next section briefly reviews the relevant theoretical aspects, such as proving tendencies in simulation models, cross-border B2C e-commerce, and oligopoly. Next, the simulation model is described. The subsequent section presents the experimental process and its design and discusses the simulation results. The final section consists of the conclusion and suggestions for future work.

2. Theoretical background

Simulation allows us to experiment virtually, leading to conclusions that can be used to understand an actual

system. This paper aims to identify and demonstrate the persistence of an emergent tendency in cross-border B2C marketplaces, specifically highlighting their oligopolistic nature using theorem-proving techniques in agent-based simulation. Consequently, this section reviews the theoretical framework regarding cross-border B2C marketplaces, themes related to our agent-based simulation approach (e.g. the endorsement mechanism used for agents' reasoning), and theorem-proving. The strengths of the theorem-proving conclusions vary depending on the model's validity and the experiment's "solidity." For instance, some researchers[15–17] developed a solid experiment with a constrained proof of tendencies in simulation outputs. Furthermore, the SocLab platform allows alternative proof forms by exploring the structure of the simulation model.[18] In addition to these ways of developing proofs, other methods of exploring the dynamic structure of a simulation exist: scenario analysis and Monte Carlo simulation, as well as theorem-proving methods. This section will briefly review all these approaches.

2.1. Cross-border B2C e-commerce: determinants and characteristics

E-commerce has experienced significant growth worldwide in recent years. E-commerce can be domestic (i.e. buyers and sellers are from the same country) or cross-border (i.e. buyers and sellers are from different countries). When companies sell to consumers through the Internet, it is called *B2C e-commerce*. Most B2C cross-border e-commerce is developed through a *marketplace*: an online market offered by a company in which buyers and sellers interact.[19]

Cross-border B2C e-commerce faces numerous uncertainties.[20–22] Common challenges for all cross-border e-commerce include the difficulty of establishing clear and confidential contracts and the absence of a well-defined and accessible international tribunal and laws to protect buyers. This is particularly problematic since buyers and sellers are often in different countries, fall under different jurisdictions, use different currencies, and speak different languages.

Previous studies about B2C e-commerce in general and cross-border B2C e-commerce in particular have analyzed drivers of and restraints on consumers when buying through B2C e-commerce. Our previous study[5] compiled the factors that determine that consumers purchase in B2C e-commerce (see Table 1). Thus, these factors were used in the proposed model.

On the individual level, previous studies have analyzed the effect of these factors on consumers intention to buy the product;[36] very few studies, however, have analyzed their on a macro-level: that is, in the total size of the market.[5] A *market* refers to the aggregate of all consumers, both existing and potential, for a specific product or service.[20] The size of a market can be measured by the total sales in the market. Each company has a portion of the total market, called the *market share*.[20] The cross-border BC2 e-commerce market comprises all consumers buying online from firms abroad; *market size* refers to the total sales. Therefore, the market share of a specific company is the proportion of sales made to consumers from foreign countries relative to its competitors.

A company's market share will depend on the strategy the company follows and on the strategy of its competitors.[23] Regarding the market share in cross-border B2C e-commerce, our previous study[5] has shown that the factors given in Table 1 determine the market share of marketplaces. The importance of each attribute for consumers in selecting which marketplace to buy from will be different. For instance, cost-saving is usually one of the most important reasons for buying a product on a foreign website.[22] Thus, if a marketplace follows a strategy of dropping prices, it can increase its market share, and the marketplace sales of competitors will decrease. As indicated above, however, several attributes are usually considered when a buyer decides on a marketplace; each attribute has a different level of importance for buyers, making the decision difficult to analyze. For modeling decision-making in this difficult context, we will use endorsements, as indicated above.

In addition, WOM—consumer recommendations about the marketplace—is also important. When consumers buy in a marketplace, they can share their experiences with others, influencing others' decisions to buy in that marketplace. WOM is considered the most important information source for consumers,[24] and it is determinant in new product diffusion.[39] Consumers usually will learn about new products, services, and shops because of the recommendations of others, and, once they know the product, they also make the decision to buy the product due to others' recommendations.[40] This is why many researchers have introduced WOM into their ABM models. For instance, Chica and Rand[41] examined the adoption of a freemium app using an ABM in which WOM was introduced; Jiang et al.[42] replicated the evolution process of market share for multiple brands competing online, as well as an ABM with WOM; Peres and Van Den Bulte[43] created an ABM to explain whether and how long it remains more profitable to be the unique seller of a product that will be introduced in the market, with WOM key to the result. In the same line as previous studies, WOM is critical to the proposed model. According to Terán et al.,[5] the key to explaining the market share of Chinese B2C marketplaces is awareness of the marketplace. The most well-known marketplace is the market leader, AliExpress, and it is very difficult for competitors to reach the leadership of the market in the medium term. WOM, however, allows consumers to learn about new marketplaces, and marketplaces

Table 1. Characteristics of B2C marketplace important to shoppers and that determine purchasing from the site.

Factor	Definition	Studies that have shown its importance in e-commerce
Reputation B2C marketplace	The extent to which buyers believe a B2C marketplace is professionally competent or honest and benevolent[23]	Teo and Liu[24]
Company size	Company overall size and market share position[23]	Teo and Liu[24]
Website appearance	Visual attractiveness of a website[25]	Liao et al.[25]
Website content quality	The characteristics of the website content in general, such as content usefulness, completeness, clarity, conciseness, and accuracy[26]	Liao et al.[25]; Ranganathan and Ganapathy[27]
Website technical quality	The extent to which appropriate technologies have been adopted by the web retailer[25]	Liao et al.[25]; Van der Merwe and Bekker[28]
Security of the web	The dependability and assurance of a retailer's online transaction system, which enables transactions through the Internet to be secure and successful[24]	Teo and Liu[24]
Privacy	The willingness of consumers to share information over the Internet that allows purchases to be concluded[29]	Belanger et al.[29];
Consumer reviews	Any positive or negative statement made by potential, actual or former customers about a product or company, which is made available to a multitude of people and institutions via the Internet[30]	Chevalier and Mayzlin[31]
Cost saving	The existence of low prices on the website	Escobar-Rodríguez and Bonsón-Fernández[32]
Product offerings	The assortment or range of goods available from a retailer[33]	Chen and Tan[34]
Product quality	The expected standard of product excellence[33]	Chen and Tan[34]
Service quality	Overall customer evaluations and judgments regarding the excellence and quality of e-service delivery in the virtual marketplace[35]	Chen and Tan[34]
Previous experience on the website	Transaction performance in previous purchases on the website	Fang et al.[36]
Previous experience with websites	Transaction performance in previous purchases on websites	Fang et al.[36]
Word of mouth	A face-to-face conversation between consumers about a product or a service experience[37]	Hwang[38]

Source: Terán et al.[5]

well valued by consumers increase their market share at the expense of other marketplaces with more awareness but poorer characteristics. Therefore, we consider introducing WOM in the model crucial to understanding the market and simulating it in a more realistic way.

2.2. Cross-border B2C e-commerce as an oligopoly

As we have explained, the transactions between agents from different countries lead to special circumstances that require separate analysis.[21] International purchases present many uncertainties (e.g. low buyers' control regarding product and service and the diversity of legal systems) that hinder international purchases.[22] These issues explain why cross-border e-commerce B2C is dominated by only a few marketplaces. This type of market is called an *oligopoly*.[44]

An oligopoly is a market structure in which a small number of firms dominate the market and control most of the market share.[44] These firms may have significant market share and influence over pricing and other market conditions.[44] As only a few firms exist in this type of market, the strategies they follow can significantly affect the market as a whole. Therefore, firms in an oligopoly are highly interdependent and must consider the strategies and reactions of their competitors when making decisions.[45]

Terán et al.[5] have shown that few companies dominate the Chinese cross-border B2C e-commerce market, which presents oligopolistic characteristics. It comprises a few competitors in marketing. This oligopoly, however, has a special characteristic that allows companies within it to exhibit monopolistic properties. One firm, AliExpress, is the clear market leader retaining most of the market share. AliExpress can set market conditions (inducing prices, post-purchase policies, and quality of product), and its strategies affect the entire market. Changes in the strategies of other e-commerce platforms will not significantly alter the market share of the companies in the medium term. Thus, AliExpress could act in the market as a monopolistic entity. At this point, our question is: What will happen in the market if we consider well-known marketplaces from other countries? The model studied in this paper addresses this question by considering a market in which Amazon is present.

2.3. Agents' reasoning: the endorsement scheme

The classic cognitive developments for representing human cognition and initial conceptual models, such as State, Operator, And Result (SOAR), Adaptive Control of

Thought–Rational (ACT-R), and Belief -Desire -Intention software model (BDI)[11,46,47] are too complex to be implemented in an Agent-Based Modelling (ABM) context. Still, simplifying the original models allows us to effectively represent the learning and evolution of the individual's mental models using the endorsement mechanism, as explained by Cohen[11] and Moss.[12] This notion has been used in a wide range of simulation models,[12,13,48–50] presenting important conceptual advances and revisions in terms of their usefulness and implications for virtual research in social interaction (e.g. to add contextualized knowledge, as diverse researchers show).[13,50,51] Among their more recent applications, Geller and Moss[50] used endorsements to represent the agents' reasoning process about preferences in models of power and authority. The authors developed a model of a characteristic agent based on interviews and indirect data, which allowed them to differentiate dimensions of reasoning and implement them in a "natural way"; that is, it permitted the use of mnemonic tokens found in the evidence, thereby overcoming the lack of statistical data. An example of an endorsement would be the "high reputation" label a buyer observes in a B2C marketplace. An extension for representing endorsements was added to the 5.2 version of NetLogo.[52,53]

2.3.1. Example of the endorsements scheme. The endorsement mechanism forms part of the rules of the ABS. It establishes how agents choose among different options. We can assume that the agents are buyers and that the choices are B2C marketplaces. Buyers would then repetitively select B2C marketplaces at each time period of the simulation. We supposed that buyers valuate the following categories of B2C marketplaces and take them into account when choosing: (1) the reputation of the marketplace, (2) product quality, (3) privacy, and (4) the security of the website. These categorical variables could take, e.g., two, three, or four levels or attributes. We can assume that they take only two values: low or high. These values define all the possible eight endorsements that marketplaces show: highReputation, lowReputation, highProductQuality, lowProductQuality, highPrivacy, LowPrivacy, highSecurityWeb, and lowSecurityWeb. We assumed that in several interactions, an agent (a buyer) has perceived in a certain B2C marketplace the following attributes: highReputation, lowQuality, highPrivacy, and highSecurity. These will determine the beliefs of the agent about the marketplace. Endorsement theory[12] tells us how to determine the importance of this B2C marketplace for the buyer by using the weight of these attributes: that is, the buyer's intention to buy in that marketplace. An important component of the endorsement mechanism is the endorsements schemata. An example of this is shown in Table 2. In Table 2, essentially, a weight is associated with each level, which can be estimated via, for instance, interviews, surveys, or data mining.

Table 2. Example of values of the attributes of an endorsements schemata.

Endorsement/attribute	Value of the endorsement for the agent
highReputation	3
lowReputation	−1
highProductQuality	2
lowProductQuality	−1
highPrivacy	1
lowPrivacy	0.5
highSecurityWeb	4
lowSecurityWeb	−2

The higher the weight of the positive case of the attribute, the greater its importance to the buyer. The formula for evaluating the endorsements and determining buyer i's indicator of shopping intention at the website j at time t (IdBuyer(i, j, t)) is as follows:[12]

$$IdBuyer(i,\ j,\ t) = \sum_{ei>0} b^{ei} - \sum_{ei<0} b^{|ei|}$$

where b is the base of the endorsement scheme, a value usually in the interval [1, 2], and ei are the values of the endorsements (beliefs) buyer i has for website j at time t given in Table 2.

To further concretize, assume that in its interaction with marketplace 4 at a certain iteration time t, agent 8 has in its record (memory) the level or endorsements highReputation, lowProductQuality, highPrivacy, and highSecurityWeb, whose weights in Table 2 are: 3, −1, 1, and 4, respectively, and that b takes the value 1.2; we then have the following indicator for buyer 8 of the option represented by the B2C marketplace 4:

$$IdBuyer(8, 4, t) = 1.2^3 - 1.2^1 + 1.2^1 + 1.2^4 = 3.8016$$

If we also assume that this buyer has experiences with two other marketplaces (2 and 5), which the buyer has endorsed, and evaluates them at time t, obtaining the following values for shopping intention IdBuyer($8, 2, t$) = 2 and IdBuyer($8, 5, t$) = 4, then buyer 8 then will choose among the three websites 4, 2, and 5 with probabilities in proportion to these IDs (i.e. with probabilities of 3.8016/9.8016, 2/9.8016, and 4/9.8016, respectively, as the sum of IdBuyers is 9.8016).

After each shopping experience, the buyers actualize the endorsements in accordance with their experiences, and could exchange endorsements (e.g. in the form of Word-of-Mouth communication).

2.4. Proving tendencies in a model

This section reviews several forms of verification and proving tendencies in simulation models.

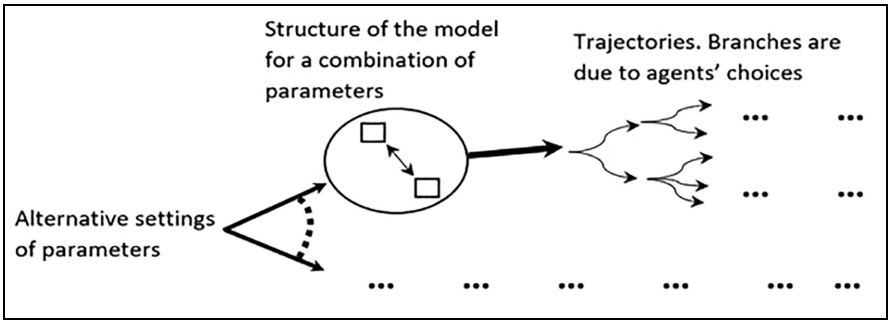

Figure 1. Exploration of the simulation model's trajectories for possible parameters and agent choices.

24.1. Constraint proof of tendencies in simulation outputs: Exhaustive exploration of the dynamics of a model. This method was applied for a simulation model of traders.[15–17,54] Simulation trajectories for the model's set of parameters and agents' choices were explored, as shown in Figure 1, which defines a subset of the dynamic structure (behavior) of the model. The model's parameters are all unique combinations of assignments of values attainable at the beginning of the simulation per the study and the experimental design; the agents' choices are all alternative decisions the actors can select at each simulation iteration time. In this sense, the exploration of the simulation trajectories is constrained to a set of values of the parameters and choices of the agents. This exploration is exhaustive because it thoroughly applies all possible values of the parameters of the model and the agent's choices defined for the exploration of simulation trajectories. A tendency, understood as a recurrent pattern, is proven if it appears in all explored branches (see Figure 1). Two criteria to evaluate the quality of the proving procedures are the proof's strengths and the pertinence of the proof for the study. This method integrates the strategies of the theorem prover OTTER[55] in the simulation process.

A constraint proof of a tendency consists of determining the necessity of the tendency in all possible simulation trajectories given by all parameter combinations of the model and all branches of each trajectory according to the possible combinations of alternative agent choices. Although it is a solid proof, it is incredibly time-consuming and difficult, considering the time constraints and programming capabilities of most research. In a previous study,[54] several authors also indicated that it was a very complex task. Generally, the number of simulation branches explored increases exponentially with the number of time iterations. Given this reality, alternative forms of proving can be performed; some of them are among the weakest forms but are nonetheless appropriate for specific studies.

24.2. Exploring the dynamic structure of the model: Proving tendencies in the SocLab platform. Alternative forms of proving tendencies in a model include the exploration of the whole dynamic structure (the possible behavior) of the model. Exploration of this kind can be developed in the SocLab platform.[18] In this case, the model's diverse parameter combinations are explored for any possibility of the agents' choices. Basic variable values, the state of the resources, or dependent-variable values, such as actor satisfaction or influence, enable the observation of tendencies in the model. For instance, the SocLab platform model's structural analysis can show that, when the satisfaction of one of two actors (either A or B) is high, the satisfaction of the other is also high; when the satisfaction of one actor is low, the satisfaction of the other is also low. These conditions imply that actors A and B are circumstantial allies, a tendency of the model; the model's structure presents this more conclusively than examining a subset of simulation trajectories. SocLab enables us to prove diverse model properties.[18] As SocLab is used to simulate systems of organized action, it allows the proof of properties in simulated organizational settings via a systematic exploration of the (finite) state space of the organization's model.

2.4.3. Monte Carlo simulation: Random exploration of simulation outputs. The most common way to explore the options of agents and other alternatives during simulation is by using probabilities, a process called Monte Carlo simulation.[56] In this case, replications of the experiment are performed, and statistical summaries about the variable values (e.g. means and variances) can be used for analysis. This proof is weaker than the two mentioned above, but it is much more appropriate and practical for this study. Its reliability increases as the number of replications increases.

2.4.4. Scenario analysis: Domain-expert exploration of parameters. Another way to explore simulation parameters is by following the interest of domain experts (people, managers, or researchers who know the systems) and/or stakeholders who define the combination of values of

interest for the parameters according to the study's focused interest. For instance, these values of interest could represent the simulated system's actual situation or define specific scenarios of interest for the study. This method allows us to validate the model by simulating it in place of the actual situation and comparing the outputs with the observation of the real system; we can also investigate for fragments of the theory represented by the model and defined by the scenarios of interest of the domain experts and stakeholders (for more about scenario analysis, see James[56] and Terán and Domingo.[57])

2.4.5. Other proving methods in simulation. The strongest form of proving tendencies in simulation is by using Theorem Provers or mimicking their methods. Theorem-proving in simulation requires applying formal methods to ensure that the outputs of the simulation models adhere to specific properties or behaviors. Theorem-proving involves the use of logical reasoning to prove that a model adheres to certain properties. Among the better known theorem provers are Isabelle[58] and OTTER.[55] This approach combines mathematical rigor and simulation process (see, for instance, the method described above in the section ''Constraint proof of tendencies in simulation outputs'' and Terán[17]). There are, however, many methods used.[9] Among these are Model Checking, a technique used to systematically explore the states of a model to verify whether certain properties hold. It is widely used in hardware and software verification; see, for instance, the tools SPIN, NuSMV, and UPPAAL.[8,9] Another related method is the Simulation-Based Verification method, which combines simulation with formal methods to cover scenarios that are difficult to simulate exhaustively. This often involves running simulations to gather data and using formal methods to prove properties about the model. Additional methods are Mathematical Proofs, Temporal Logic Abstract Interpretation, and Falsification.[8,9,58] Applications of all these methods range from simulation models of agents to aerospace, automotive, and critical software systems.

2.4.6. Difficulties for theorem-proving and proving tendencies in simulation. Theorem-proving in simulation is highly valuable but presents several challenges and weaknesses, including:

(1) The complexity of models due to state space explosion. As the number of parameters and alternatives of these increase, the complexity of exploration of simulation trajectories augments (i.e. the number of potential explorable states) grows exponentially, making it difficult and many times infeasible to explore all potential states;[8,9,54]
(2) The limitations of formalism (expressiveness vs decidability). The more expressive the formal languages are, the wider the range of properties they can describe; this increased expressiveness, however, often makes them undecidable;[9]
(3) Abstraction challenges. Creating abstract models that accurately represent the real system while being simpler to analyze is challenging. Inaccurate abstractions can lead to incorrect conclusions about the system's properties;[8]
(4) Scalability issues and resource constraints. Theorem-proving and model-checking require substantial computational resources as the size of the model grows, as does the demand for processing power and memory;[8]
(5) Tool limitations. Existing tools have limitations in terms of scalability, usability, and integration with other simulation tools. This can hinder their practical application in complex systems;[8,9]
(6) Human factors. Theorem-proving and formal verification require a high level of expertise in formal methods, including logic and mathematics;[8]
(7) Real-world applicability and realism. Sometimes it is difficult to ensure that simulations accurately reflect real-world systems;
(8) Dynamic and adaptive systems. As some domains systems evolve, verifying their properties requires handling changes and adaptations.[8]

3. Methodology: design and application of an ABM

This section presents in its design a model of B2C marketplaces. ''Survey'' describes the survey used to build the model. ''ODD description of the simulation model'' then describes the model using the ODD (Overview, Design concepts, and Details) protocol.[59]

3.1. Survey

We developed a survey to obtain the necessary data to build the model. We obtained two types of data with the survey. On one hand, we ascertained consumers' perceived importance of the website characteristics that determine their purchases in B2C marketplaces (as presented in Table 1), and, on the other hand, we obtained consumers' perceptions regarding the extent to which certain B2C marketplaces have these characteristics. The participants were individuals who had previous shopping experience with or were familiar with B2C marketplaces. To this end, a questionnaire based on previous work was constructed, thereby maintaining the correspondence between existing theories and our research results.[60] Characteristics of B2C marketplaces in relation to purchasing were obtained from previous studies (see Table 1), and scales were adapted from the literature to measure them (see Table 3). To

Table 3. Table displays for each variable and the items for (1) measuring the characteristics of a B2C marketplace and (2) measuring consumers' perceptions of a B2C marketplace.

Variable[1]	Measurement of characteristics of a B2C marketplace Items and Source	Measurement of consumers' perceptions of a B2C marketplace
Reputation of the B2C marketplace	• The company is well known. • The company has a good reputation in the market. • The company has a reputation for being honest. • The company has a reputation for being fair. • The company has a reputation for being consumer-oriented. Source: Sen and Lerman[37]	• Company reputation (1 = bad reputation, 7 = good reputation) Source: Self-developed
Company size	• The company is very large. • The company is one of the industry's biggest suppliers on the web. Source: Hwang[38]	• Company size (1 = small, 7 = large) Source: Self-developed
Website appearance	• Website attractiveness • Website organization Source: Libai et al.[39]	• Website attractiveness (1 = little attractive, 7 = very attractive) • Website organization (1 = little organized, 7 = very organized) Source: Libai et al.[39]
Website content quality	• Finding contact information • Finding company information • Finding product details • Finding customer policies • Finding customer support • Presence of visual presentation aids (graphics, audio, video) Source: Libai et al.[39]	• Website content quality (1 = low quality, 7 = high quality) Source: Self-developed
Website technical quality	• Ease of navigation • Search facilities • Absence of broken links • Absence of "under-construction" pages Source: Libai et al.[39] and Chica and Rand[41]	• Functioning of the website (1 = bad functioning, 7 = good functioning) Source: Self-developed
Security of the website	• Availability of secure modes for transmitting information • Provision for alternative, non-online modes for financial transactions • Opportunity to create individual accounts with log-on ID and password • Provision for PayPal payment • Website provides a security seal Source: López and Sicilia[40]	• Security of the web at buying (1 = little security, 7 = high security) Source: Self-developed
Privacy	• Gathering of personal information Source: López and Sicilia[40]	• Privacy at gathering of personal information (1 = low privacy, 7 = high privacy) Source: López and Sicilia[40]
Consumer reviews	• Provision of star ratings of the products • Provision of consumer reviews of the products • Provision of consumer reviews of the vendors Source: Self-developed	
Cost saving	• Obtaining the best prices for products • Getting a better price–quality relation • Saving money Source: Jiang et al.[42]	• Product pricing (1 = expensive, 7 = cheap) Source: Self-developed
Product offering	• Great range of products on the website • Finding hard-to-find products on the website Source: Peres and Van Den Bulte[43]	• Variety of offered products (1 = little variety, 7 = high variety) Source: Self-developed
Product quality	• Finding quality products Source: Peres and Van Den Bulte[43]	• Quality products (1 = low quality, 7 = high quality) Source: Peres and Van Den Bulte[43]

(continued)

Table 3. Continued

Variable[1]	Measurement of characteristics of a B2C marketplace Items and Source	Measurement of consumers' perceptions of a B2C marketplace
Service quality	• Speed of delivery • Ease of returning merchandise Source: Kotler et al.[44]	• The speed of delivery (1 = slow, 7 = fast) • The ease of returning merchandise (1 = difficult, 7 = easy) Source: Kotler et al.[44]
Previous experience on the website	• Previous experience purchasing on the website Source: Self-developed	• Previous experience purchasing on the website (satisfaction) Source: Self-developed
Word of mouth	• Friends and relatives' experience purchasing on websites Source: Self-developed	

Source: Terán et al.[5]

avoid the problem of long questionnaires so as to increase modal responses and decrease extreme responses, which can affect the quality and validity of the data,[61] we used a small number of items to evaluate each website (generally one item per characteristic; see Table 3). We used 7-point scales to measure the concepts.

The marketplaces evaluated were identified in a pretest. They are Amazon, AliExpress, LightInTheBox, MiniInTheBox, DealExtreme, Banggood, DHgate, Delfind, Wish, and Alibaba. Participants were asked to give the names of and evaluate any other marketplaces they knew and had bought from, stating how often they bought from them, how many other users they had frequent contact with (to measure their strong ties), and how often they shared information with them. Finally, their sociodemographic information was obtained.

The population consisted of Spanish Internet users. To obtain a sample, a market research firm with an online consumer panel was contracted. The firm distributed the questionnaire link to individuals registered in its online panel. The company assured us that the sample fulfilled the characteristics of Spanish Internet users in terms of sex, age, and place of residence. These measures were taken to ensure the representativeness of the sample.[62] The final sample consisted of 414 individuals.

3.2. ODD description of the simulation model

We explained our ABM for this study using the ODD (Overview, Design concepts, and Details) protocol, a widely known protocol used to describe ABM models.[59] As Figure 2 shows, this model is composed of three main categories: Overview, Design concepts, and Details. Each category describes a different aspect of the model. The Overview describes elements like the purpose and variables in the model. The Design concepts category describes how agents interact in the model. The Details category describes

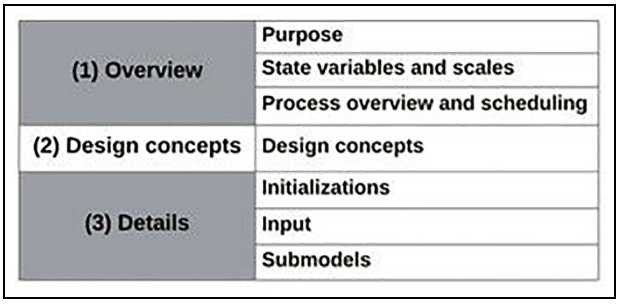

Figure 2. Description of the ODD protocol.

the initialization and calibration of the models. This section describes our model using these three categories.

Availability. This model was implemented in the Java programming language used in a previous study;[5] the program can be downloaded from a public repository on GitHub.[14]

3.2.1. Overview. This section comprises three parts: (1) purpose and patterns; (2) entities, state variables, and scales; and (3) process overview and scheduling. The subsections describe these three elements in more detail.

***Purpose and patterns*:** The research and model address the problem of modeling and supporting decision-making in cross-border B2C marketplaces. The model represents buyers' shopping (interactions between buyers and marketplaces and between buyers via WOM), which allows us to observe the performance of the marketplaces in question, measured as the number of buyers purchasing at least once in a marketplace. This informs and supports the stakeholders' decision-making by modeling a specific case: cross-border B2C e-commerce in Spain. The information used in the simulation was obtained from a previous survey (see Terán et al.[5]).

***Entities, state variables, and scales*:** The environment consists of a set of seven (computational) agents representing the marketplace type and a set of *m* (computational) agents representing the buyer type (online shoppers). The six better-known marketplaces (Amazon, AliExpress, LightInTheBox, Banggood, Wish, and Alibaba) were selected, along with ''Others,'' agents representing the remaining lesser-known marketplaces (i.e. those known by fewer than 10% of buyers). Table 7 of Appendix A shows the percentage of shoppers familiar with these marketplaces.

In the simulation, buyers learn from their experience via endorsements (as they observe the characteristics of the marketplaces, such as the quality of the product) and adapt their behavior during the simulation. Marketplaces do not learn but maintain similar behavior throughout the simulation. Endorsements involve retrieving the accumulated data representing buyers' experience (beliefs) with the marketplaces. The following paragraphs explain how marketplaces' attributes and buyers' endorsements are determined.

The marketplaces' attributes were obtained from previous studies on e-commerce, as described in the previous section. We assumed that each marketplace attribute has two possible states only: high and low. The probability of each possible state was calculated from the survey based on the frequency of an attribute receiving low or high valuations from the interviewee. Thus, the frequency for ''low'' was obtained by counting the number of answers in the survey in the interval [1, 4] (half of the whole interval: (1, 7)). The probability of a low valuation was calculated by dividing this frequency by the total number of answers. The probability of a high valuation is the complement of the probability of a low one. These values are presented in Table 9 of Appendix A. Given that we have 12 categorical variables with two possible states each (high and low), a marketplace's total number of possible attributes is 24 (e.g. highReputation, lowReputation). For example, supposing that in several interactions, an agent has perceived the following specific attributes about a particular marketplace: highReputation, lowQuality, highPrivacy, and highSecurity. These will determine the beliefs of the agent about the marketplace. The endorsement theory (Moss 1995) tells us how to determine the importance of this marketplace for the buyer using the weight of these attributes—that is, the buyer's intention to buy in that marketplace. The survey obtained these weights as a function of the importance of the marketplace's attributes for the interviewed buyers (see Table 8 of Appendix A). The endorsements were calculated as follows: each endorsed attribute can take a high value (considered positive or satisfactory) or a low value (considered negative or dissatisfactory) according to how it appears to the buyer. Simon's study[63] indicated that agent satisfaction is usually in a range of values above zero and that dissatisfaction is in a range below zero: ''Most people consistently register either slightly below zero (mild discontent) or a little above (moderate satisfaction)'' (p. 29). For each attribute the buyer takes into account (e.g. product quality), we calculated the mean value of its importance to the buyer using data from the survey (Tables 8 and 9, Appendix A). The range of the answers in the survey is (1, 7) with 4 being the neutral central value. This neutral value corresponds to the weight of the endorsement of 0. Consequently, the range of weight of the endorsements is (–3, 3) the maximum value of the weight given to an endorsement is 3, and the minimum value is –3. The value of a positive endorsement for an attribute is defined as the distance from the mean of the attribute to the central value: for example, as the mean of the importance of ''product quality'' is 6.25, the positive endorsement (for a satisfactory experience: high-product quality) is established as $6.25 - 4 = 2.25$. Following Simon's[63] suggestion that a dissatisfactory experience is represented by values lower than those of a satisfactory experience, the corresponding value for a negative experience is taken as the negative value of one half of the value of the satisfactory case; for example, in the event of low-product quality, the endorsement is $-2.25/2 = -1.125$.

Process overview and scheduling: The iteration period is the time between two sequential shopping instances (Figure 3). The simulation process implements the following features of the system according to the overall parameters of the simulation, as shown in Table 10 of Appendix A.

(a) *The initial state and structure of the system*. This involves creating the agents with their variables, including the two values, low and high, of the 12 possible attributes of the marketplaces (Table 9, Appendix A) and the buyers' endorsement scheme.

(b) *The simulation dynamic for the iteration steps (time iterations).* The simulation was run for each iteration, from 1 to *n*, where *n* is the maximum number of iterations. Usually, *n* was set to 200, but results were collected for iterations 101–150, the points at which the buyers exhibit a learning behavior and the stationary state is reached after the transient state. At this stage, we have the agents' interaction at each step of the simulation: sellers showing their attributes (such as quality) and buyers' shopping and endorsing (learning). We estimate that a complete shopping experience lasts approximately 2 months. The span of 50 iterations, during which results are collected (from iterations 101 to 150), corresponds to roughly 8 years in real-time. For iterations 1 to *n*, we have the following:

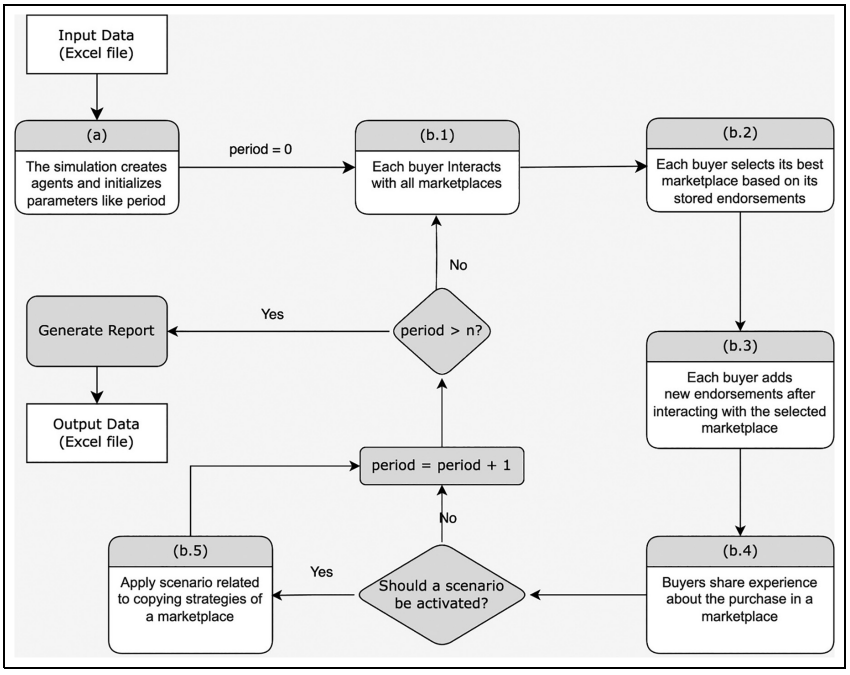

Figure 3. Process overview of the ABM used in the study.

(b.1) *Actions of the marketplaces.* At the actual iteration time, each marketplace randomly presents the levels (states) of the 12 categorical variables indicated in the left column of Table 9 in Appendix A, which represent the characteristics important to the buyer's choice. (The section ''Entities, state variables, and scales'' explains how these values were obtained).

(b.2) *Actions of buyers.* At the actual iteration time, a buyer uses the endorsement scheme (defined as the importance of attributes for buyers shown in Table 8, Appendix A) to calculate the endorsement value for each marketplace with which the buyer is familiar; the buyer then probabilistically chooses a marketplace according to these endorsement values (i.e., the probability weights for each marketplace).

(b.3) *xAddition of new endorsements.* After purchasing from the selected marketplace, the buyer endorses it by adding to the list of attribute-related endorsements the marketplace shows in the present iteration, thereby updating its memory with the new experience. While the simulation of this model advances (i.e., increases the period), each buyer collects a longer set of endorsements, suggesting that the decision to select a marketplace can vary according to current endorsements.

(b.4) *Buyers share their experiences.* Finally, to simulate WOM in the Spanish context (as explained below), the buyer shares (12 buyers chosen at random) the name and attributes of the marketplace they know best (i.e., the one with the highest endorsement value) with other buyers. Because of bounded rationality, from all the information a buyer receives from the 12 other buyers, only one suggested seller in the next iteration is considered (chosen randomly). The section ''Entities, state variables, and scales'' explains how the values for the endorsement scheme were determined. To test model robustness, we ran simulations with the WOM feature disabled (see the section ''Relative importance of the factors driving the tendency'').

(b.5) *Application of a scenario.* Once the agents have learned the behavior of the actual system, the system's structure is varied to represent the scenario. A scenario is used to analyze a strategy of interest for a particular marketplace. The analyzed scenarios are described in the ''Scenario analysis'' section.

3.2.2. Design concepts. This section describes the relationship between the model and the areas of human social behavior considered necessary for the social-simulation community according to the ODD protocol.

***Emergence and learning*:** The number of actors does not change during the simulation. Emergence occurs during the buyers' decision-making activities (endorsements), as it evolves according to their preferences and the marketplaces' attributes, improving the buyers' behavior. The overall marketplace behavior is shown by the market share, in which we consider the number of buyers who have bought at least once in a particular marketplace (SalesUniquePerMarket). The tendency of this variable

demonstrates the performance of the marketplaces, the outstanding behavior of which can be considered surprising or emergent by the stakeholders and researchers. The (implicit) objective of the buyers while learning is to increase satisfaction by buying in the marketplace whose characteristics are closest to their preferences, as represented by the weights they give to the diverse attributes of the marketplaces according to the endorsement scheme's levels. Adaptation and objectives, however, are not explicitly included in the model. They occur implicitly as buyers learn via the endorsement scheme.

***Prediction*:** There is no prediction in the model. The model is useful for explaining but not predicting.

***Sensing*:** Buyers realize the attributes of the marketplaces when they interact with them while shopping. Marketplaces do not ''sense'' their environment. Their behavior changes stochastically during the simulation as they vary the attributes of their website from iteration to iteration. Buyers communicate with each other via WOM.

Interaction. Buyers interact with each marketplace. During each shopping interaction and at each time step, the buyers endorse the marketplace according to the attributes shown, and the marketplaces present their attributes stochastically. Interaction between buyers occurs via WOM.

Stochasticity. Stochasticity is implemented when marketplaces present their attributes according to their frequency in the survey and when buyers choose between marketplaces according to the index generated by the endorsements' function.[5,12] We assume a uniform distribution.

***Collectives*:** There is no grouping in the model. Buyers and marketplaces act individually.

***Observation*:** The variable observed in the simulation represents the number of buyers who bought at least once in a particular marketplace (SalesUniquePerMarket). We consider this measure an approximation of the marketplace's market share. This variable was observed for each time step; a mean was then calculated for each marketplace at each iteration time step for each of the 30 replications of the experiment.

3.2.3. Details. This section explains three details of the model. First, we describe the necessary configuration to create a set of simulations with the same aspects. Second, the input data required to start the simulation is explained. Finally, we briefly describe how a buyer interacts with a marketplace.

***Initialization*:** The initialization includes the number of users (buyers) and their friends, the number of periods in each simulation and number of simulations, the number of learning-process periods, the attributes of the marketplace, and the endorsement variables representing information about previous transactions that the agents could retain (i.e. experience for learning). The initialization was loaded in the simulation software using Excel.

***Input data*:** The data include the number of times buyers buy in a marketplace, the weight of the buyers' endorsements, and the probabilities of the marketplaces' attributes. For each period and buyer, all values were saved in an Excel file, which we analyzed according to scenario conditions. When the simulation ended, the file contained the complete history of each buyer's data in the simulation for the variables of interest.

***Submodels*:** The entire process is explained in the ''Process overview and scheduling'' section. To briefly summarize, however, the section describes how buyers decide to buy in a specific marketplace for each of the simulation's periods and then make the transaction. To reach a decision, agents receive information from friends (other agents) and evaluate their historical transactions in each marketplace using endorsement attributes (Table 9 of Appendix A). Finally, buyers select the best marketplace evaluation to share with their contacts at each iteration step.

4. Scenario analysis: proving the oligopolistic character of the tendency

This section describes the experimental design and results for a sequence of experiments aiming to (1) identify and prove a tendency and (2) determine the relative importance of the factors generating the tendency. The section is organized as follows. First, we describe how we explored the parameters and agents' choices in the model and compare this exploration of simulation trajectories with the constraint exploration described in the section ''Constraint proof of tendencies in simulation outputs.'' Second, we outline the application of our method to identify and prove the tendency that the cross-border B2C e-commerce market has an oligopolist character. Third, we provide an additional explanation of the model described above and an analysis of the scenarios determining the relative importance of the factors that drive market share and the observed tendency. Finally, we describe some characteristics of our proving method, including its advantages and disadvantages, in light of the existing literature (see the section ''Theoretical background'').

The model was validated against data from the survey. Below, we define several scenarios, with Scenario 0 representing a base case defined as the real case: i.e. the validated case. The validation was performed following a procedure similar to that given by Leger.[14] An example of the output for Scenario 0 (with 300 buyers and 30 replications) can be found at https://tinyurl.com/ABM-Ecommmerce-baserun.

4.1. Exploration of the parameters of the model and the agents' choices: scenario analysis and the Monte Carlo method

Our exploration of simulation trajectories (Figure 1) is based on scenario analysis and the Monte Carlo method.

The domain experts (in our case, the researchers developing the study) defined the parameters, and the agents' options were probabilistically determined. For each parameter definition, replications of the simulations were implemented to explore the alternative values of the agents' choices and the mean of the examined variables of interest. Both reduce the complexity of the exploration of simulation trajectories compared with constraint proofs of tendencies in the simulation outputs (see the section "Constraint proof of tendencies in simulation outputs"). This exploration of simulation trajectories permitted us to determine the factors driving the market share. In the following sections, we will identify the diversity of parameters and the choice of the model.

Tables 7–10 in Appendix A present the entire set of parameters: (1) the percentage of buyers familiar with each marketplace; (2) the mean of the importance of the attributes for buyers; (3) the mean low and high values of the attributes for the marketplaces, as perceived by the buyers; and (4) the parameters defining the overall configuration of the model. Most of these values, specifically those illustrated in Tables 7–9 (Appendix A), come from the survey given to the marketplace users.[5] Table 10 shows the parameters used for the configuration of the model: e.g. the number of buyers, the number of simulation runs, whether WOM is activated, and the base of the endorsements schemata; some of the parameters, such as the base of the endorsement schemata, were determined during the calibration of the model.

Given the study's aim (to determine the factors driving market share) and the corresponding advice of the domain experts, variations in the parameters shown in Tables 8 and 10 were discarded, with the exception of whether WOM is activated (Table 10, Appendix A). Regarding the parameters in Tables 7 and 9 (Appendix A), the domain experts suggested specific variations as a set of scenarios defined below.

As stated, the number of parameter values was constrained by following the advice of the domain expert, and the number of agents' choices was constrained by randomly exploring some of these choices. To understand what the reduction of complexity of the exploration of simulation trajectories allows, we can compare the number of simulation trajectories in the case of doing this reduction with a constraint exploration, assuming that we have nP parameters, each of which has oPi, i = 1, 2,..., nP possible values (at the beginning of the simulation) and nA agents at each iteration time with oAj, j = 1, 2,..., nA options; if we also simulate nL time iterations and, in the case of our exploration of the simulation trajectories, perform nR replications, then the number of branches will be:

- In our study: $\prod$ i (oPi) * nR = (3*3)*(30) = 270 trajectories.
- Constraint exploration of simulation trajectories: $\prod$ i (oPi) * ($\prod$ j (oAj) * nL).

The number of replications (30 or 60) is usually much lower than the number of time iterations (100 or 1000). We can see that the second exploration of simulation trajectories is much larger and resource-consuming than ours.

To illustrate this, we can assume that we have three parameters and three options for each one and three agents and three options per agent; we then simulate 100 time iterations and, for our practical exploration of simulation trajectories, develop 30 replications. The number of simulation branches subsequently explored in each case would be:

- In our study: $\prod$ i (oPi) * nL = (3*3)*(30) = 270 trajectories.
- Constraint exploration of simulation trajectories: $\prod$ i (oPi) * ($\prod$ j (oAj) * nL) = 3^3* 3^3* 100 = 72,900.

If the number of agents increases—to 5, for instance—in the second case, the number of branches will increase to:

$$\prod_i(\text{oPi}) * [\prod_j(\text{oAj}) * \text{nL}] = 3^3 * 3^5 * 100 = 656,100.$$

Considering a more extreme case, if we have three parameters with 10 options, 10 agents with three options, and 100 iterations, the number of branches to be explored is: (10^3) * [10^3* 100] = 10^8 = 10,000,000. We would have to explore 10 million branches. If the number of actors increased to 100, the number of simulation branches explored would increase to 100,000,000,000.

The number of branches in the constraint exploration of simulation trajectories grows with the number of parameters (nP) (exponentially), the number of agents (nA) (exponentially), and—though less importantly—linearly, with the number of iterations (nL). Still, our exploration of a simulation trajectories proposal (in this paper) only grew exponentially with the number of model parameters. In the first case, these facts strongly affect the viability of research because of the involved technical issues and the time and memory capacities required. We suggest using the method specified above to constrain the number of explored simulation branches, maintaining the quality of the proof and its conclusions by further exploring specific parameters (assisted by domain experts) and agent choices (as determined by an appropriate researchers' strategy for directing the exploration of agents' options), as explained below.

4.2. Developing a practical proof

We elaborated a practical proof whose process is as follows:

(1) Identifying the tendency. The structure of the model is explored by following the procedure described above to identify a tendency.

Table 4. First set of scenarios. Scenario 0 (base model) and three scenarios simulating AliExpress's attempts to overcome Amazon's dominance in Scenario 0.

Scenario	Factor AliExpress copies or imitates from Amazon	Survey items defining the imitated factor
Scenario 0:	Base model	None
Scenario 1:	Quality and product offering	Product offering, product quality, and service quality
Scenario 2:	Web page attributes	Website appearance, website content quality, website technical quality, security of the web, and privacy
Scenario 3:	Concatenation or union of scenarios 1 and 2: quality, product offering, and web page attributes	Product offering, product quality, service quality, website appearance, website content quality, website technical quality, security of the web, and privacy

(2) Verifying the persistence of the tendency found in step (1). We verify the strengths of the previous conclusion (the tendency) by confirming the persistence of the tendency in the structure of the model through additional exploration of simulation branches beyond the simulation branches explored in step (1) to generalize or prove it (because the confirmation could be a success or a failure). For this, additional exploration of the simulation branches is done, varying the number of agent choices considered and agents in the model, to look for the tendency, which, if successful, allows us to prove it. This additional exploration requires that:
   (a) The parameters of the model are not varied. We are only interested in Scenario 0, which represents the actual system.
   (b) The number of experiment replications is significantly varied. The number of replications is increased in relation to the previous statistical exploration (done to identify the tendency). This is important given that our exploration (1) is not exhaustive (since it is in a constraint proof, as explained in section "Constraint proof of tendencies in simulation outputs"). Our model consists of a broader exploration of agent choices.
   (c) Increasing the number of actors. In general, the number of actors can be varied (either increased or decreased), but, given that the number of actors in our case study (the Spanish market) is higher than that in the model, we are particularly interested in increasing the number of actors. This will indicate whether, when the model is enlarged, the tendency maintains.

If the tendency does not present variations for the results of experiments (b) and (c)—i.e. if the exploration is successful—we will have proven that the tendency holds for the theory behind the model beyond the initial identification given in step (1). The model is not affected by variations in the number of replications or the number of agents. This demonstrates the robustness of our results, which considerably strengthens our conclusions about the observed tendency, as this is an alternative to the constraint proof presented in section "Constraint proof of tendencies in simulation outputs." Our proof is more indirect but effective because it is much less resource-consuming and more practical than the proof described in that section.

In the following subsections, this procedure is applied to our case study.

4.2.1. Identifying the tendency: Oligopolistic character of the cross-border B2C e-commerce market. Our first paper,[5] in a model similar to the one we present here but that excluded Amazon, found that the leading marketplaces in terms of the variable *unique sales* (or market share) were AliExpress and Wish (as noted, Amazon was not simulated, but the model was successfully validated in that study). The authors later examined scenarios where the strategies entailed Wish attempting to overcome AliExpress's highest value of unique sales. The results showed AliExpress's solid dominance as Wish tried to become the market leader in the measured variable (*unique sales*). The study proved this trend in a sensibility analysis with certain input variations in the simulation, specifically with changes of 5% and 10% (both increases and decreases) in the number of buyers familiar with each marketplace. A similar result was found in this study, which did include Amazon in the model.

Given that the pattern of the market repeats itself in our model (Figure 4), with Amazon becoming the dominant marketplace, followed by AliExpress and Wish, we used similar scenarios to those found in Terán et al.' study[5] to test Amazon's dominance. These scenarios are presented in Table 4. In these scenarios, the strategy is implemented by AliExpress rather than Wish to determine whether this firm could overcome Amazon's market dominance.

Figure 4 presents the simulation of the base model or Scenario 0 and the other scenarios. The results show

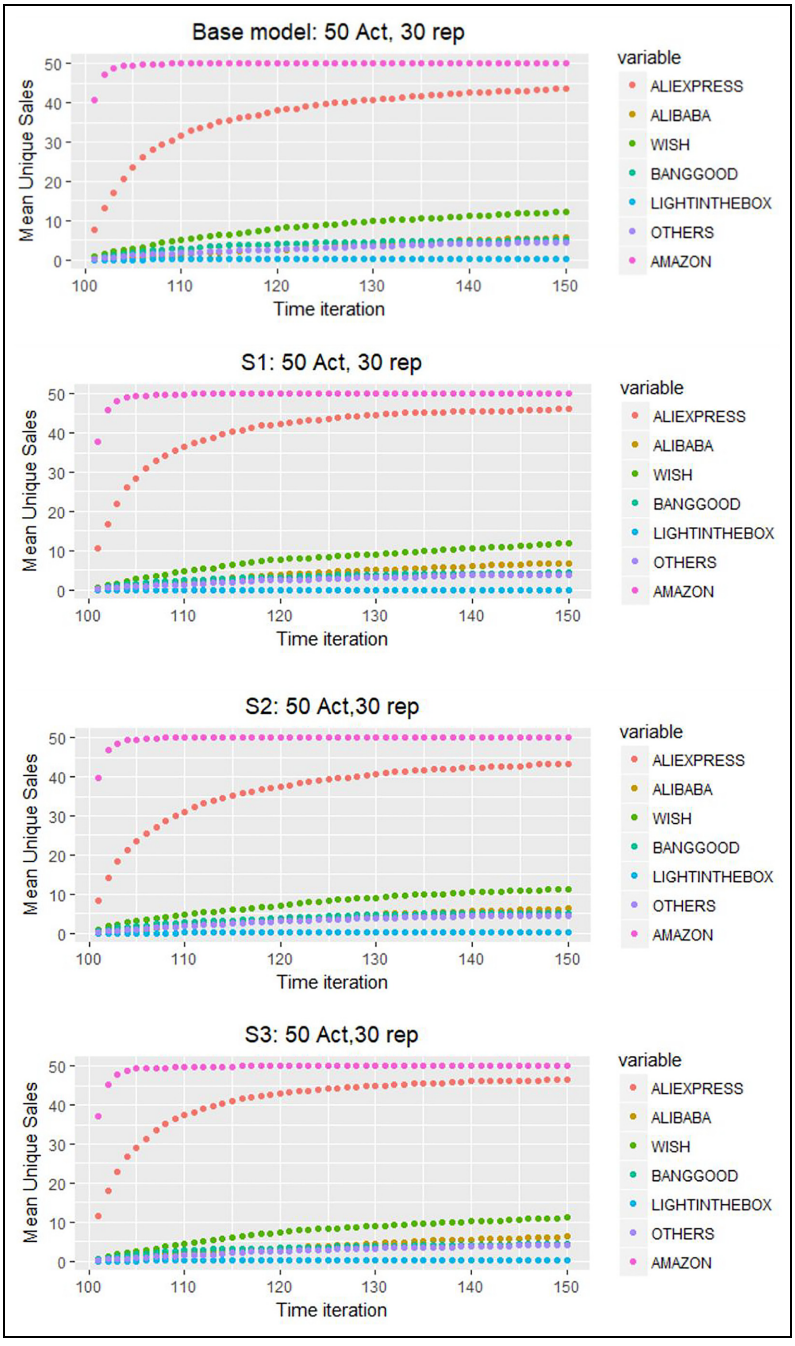

Figure 4. Base model (Scenario 0) and Scenarios 1–3. In Scenarios S1–S3, AliExpress imitates Amazon's behavior by copying the following attributes: S1: quality and product offering; S2: webpage attributes; S3: the union of S1 and S2 (quality, product offering, and webpage attributes).

Amazon's market dominance, followed by AliExpress and Wish in terms of the variable *unique sales*, a form of measuring market share. Scenarios 1–3 show that AliExpress's strategies—copying Amazon's attributes to outperform the company—are unsuccessful. This result is similar to the one that Terán et al.[5] found, with Amazon excluded from the simulation, showing AliExpress as the dominant marketplace. In a similar simulation, Wish failed to improve its performance or outperform AliExpress when copying the latter company's attributes.

Thus, in both our initial paper[5]and our results here, the dominance of one or two companies over the others is observed in unique sales, with the first company maintaining its dominance despite the strategic improvement of the

Table 5. Parameters varied to verify the validity understood as scalability of the model.

Varied parameter	Original value in the base model	New values given
Number of buyers	50	100, 200, 300
Number of simulation replications	30	100

marketplace in second place. This suggests that the cross-border B2C e-commerce market is an oligopoly. To verify the strength of this conclusion, in a practical proof, several model parameters are varied in the next section to check whether this tendency holds under such variations.

4.2.2. A practical proof: Verifying the generality of the tendency and robustness of the results. The *number of replications* and the *number of actors* were significantly increased in the experiment, as shown in Table 5. These two parameters are part of the complete set of model parameters presented in Appendix A and particularly the subset shown in Table 10.

Figure 5 presents the final values of the results for the output variable *means of unique sales* in the following models indicated in the horizontal edge in the general form "i–j," where i and j represent the following:

- i = 1: original model with 50 actors and 30 replications.
- 2: number of replications is increased to 100; thus, results correspond to a model with 50 agents and 100 replications.
- 3: number of agents is increased to 300; thus, results correspond to a model with 300 actors and 30 replications.
- j = Scenario j, shown in Table 5.

Other configurations not shown as part of this study (e.g. the number of agents: 100 and 200) were carried out with similar results to those presented here.

Finally, the last result on the right side of the horizontal edge in Figure 5 indicates the values for Scenario 0 without WOM, which is used in the analysis presented in the next subsection.

As seen in Figure 5, results do not change for the variations of the values of the factors—i.e., the model is indifferent to variations in the number of replications and agents. This proves the persistence of the tendency independent of the explored agent choices; that is, the tendency would still occur if we performed an exhaustive exploration of agent choices (as suggested in the section "Constraint proof of tendencies in simulation outputs"). Unlike the one described above, our proof is indirect. No variation regarding the tendency was found when the number of agents increased, which shows that the model is valid (scalability) beyond the configuration (50 agents), whose validity against real data was also demonstrated (see the first part of section 4). To summarize, we deduced that the tendency holds regardless of the number of agent choices explored and that the model is valid even if the number of agents increases. Therefore, we have indirectly proven the following tendency: the cross-border B2C e-commerce market shows oligopolistic behavior and is dominated by two marketplaces, Amazon and AliExpress; this tendency occurs regardless of the number of agent choices explored or the number of agents simulated in the model. The variation in the number of replications permitted us to verify the robustness of our conclusion (the observed tendency) when increasing the explored agents' choices. The *robustness of a conclusion in simulation* refers to the degree of confidence that the conclusions drawn from simulation results are valid, reliable, and applicable under a variety of conditions.

Finally, the (general) *robustness of a simulation model* refers to the ability of the simulation model to produce reliable and correct results even when faced with uncertainties and variations in its parameters. The robustness of our model is shown by the validation of our model (as indicated in the first part of section 4), along with the consistency of the results throughout a wide exploration of parameters in the model, the number of agents (scalability), and agent choices under a variation in the number of replications.

The first index in the horizontal line at the top indicates 2-tuple or couple pairings (i.e. the number of agents and number of replications), simulated as follows: 1: (50, 30), 2: (50, 100), 3: (300, 30); the second index indicates the number of the scenario, as defined above (e.g. 0: base model).

4.3. Relative importance of the factors driving the tendency

The previous subsections explain why the cross-border B2C e-commerce market is oligopolistic and highlight the difficulties in changing this structure; even when employing specific imitative strategies, it is difficult for the marketplace in second place to overtake the strongest, most dominant one. To better understand the complexity affecting the market's structure, we investigated the relative importance of three factors influencing market share: (1) how well-known the marketplace is, (2) marketplace attributes, and (3) WOM. To do this, we used the results in the previous sections to create and define three new scenarios.

First, we evaluated the effect of WOM. The results illustrated on the right side of the bottom horizontal edge of Figure 5 (NW-0) correspond to the base model with WOM suppressed. These results are very similar to the outputs of simulations of the base model with WOM (horizontal edge points 1-0, 2-0, and 3-0). This similarity shows that the results are not affected by WOM, so

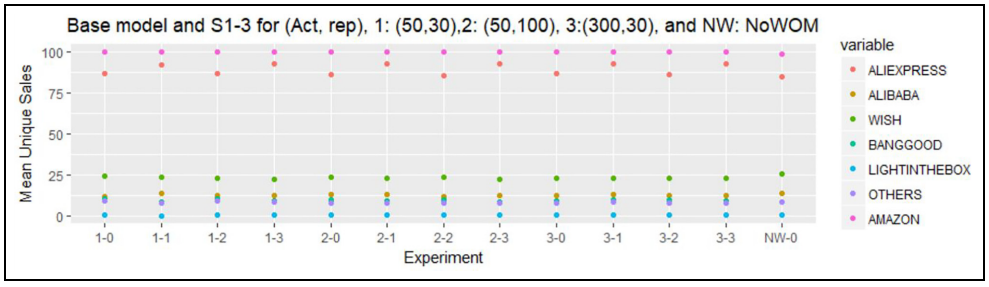

Figure 5. Mean of unique sales for several scenarios by varying the number of agents and replications. Similar to the previous experiments, data were collected over 50 iterations in all cases, from iteration 101 to iteration 150 (which corresponds to approximately 8 years in real time).

Table 6. Second set of scenarios. The new parameter values of the model for Scenarios 4–6 are indicated in the third column.

Scenario	Varied factor	Value taken by the factor
Scenario 0: Base model		
Scenario 4: Banggood is as well-known (%) as Amazon	Percentage of buyers familiar with Banggood, shown in Table 7 (Appendix A) (replaced by the percentage of buyers familiar with Amazon)	Set to 92.27 (original value: 10.14)
Scenario 5: LightInTheBox has Amazon's attributes	LightInTheBox's attributes probability, shown in Table 9 (Appendix A)	Columns (in the Excel input file) for LightInTheBox are replaced by the columns for Amazon
Scenario 6: Amazon takes LightInTheBox's attributes	Amazon's attributes probability, shown in Table 9 (Appendix A)	Columns for LightInTheBox are replaced by the columns for Amazon

variations in this factor very rarely affect the structure of the simulation model or its oligopolistic character. This conclusion is similar to the one drawn by Terán et al.,[5] whose study excluded Amazon. After considering this conclusion, our focus was on observing how the structure of the simulation model was affected by the following factors: (1) how well-known the marketplace is and (2) marketplace attributes. To do this, we created three new scenarios (Table 6), the results of which can be found in Figures 6 and 7. The justifications for choosing these scenarios were as follows:

- Scenario 4 (Banggood is as well-known as Amazon) permitted us to observe the effect of increasing how well-known the marketplace is on the performance (unique sales) of a marketplace with good attributes but that is not well-known at present.
- Scenario 5 (LightInTheBox has Amazon's attributes) allowed us to evaluate the effect of good attributes on a marketplace with poor performance (LightInTheBox).
- Scenario 6 (Amazon has LightInTheBox's attributes) enabled us to evaluate the effect of negative attributes on a marketplace with good performance (Amazon).

The results of these scenarios were used to verify which parameter—either (1) ''how well-known the marketplace is'' or (2) ''attributes''—has a more substantial effect on market share (unique sales) and is more responsible for the tendency found. The results are as follows (Figure 6):

- Scenario 4: Graph (b) shows that Banggood strongly increases the market size compared with Graph (a). Being widely known (at Amazon's level) is a key factor in improving performance (''being known'' is an important factor).
- Scenario 5: Graph (c) illustrates how LightInTheBox's unique sales remain close to those of other underperformers, similar to Graph (a) despite having the best attributes in the market (taken from Amazon). The resulting poor improvement of LightInTheBox suggests that ''attributes'' are less important than ''how well-known the marketplace is.''
- Scenario 6: Graph (d) shows that, even with the worst set of attributes, like those of LightInTheBox, Amazon's performance is not poor; its market share is lower than in other scenarios, but it only drops to second place. Once again, the parameter ''how well-known the marketplace is'' shows that this parameter is critical.

All these results are better illustrated, in terms of the simulation's final values, in Figure 7.

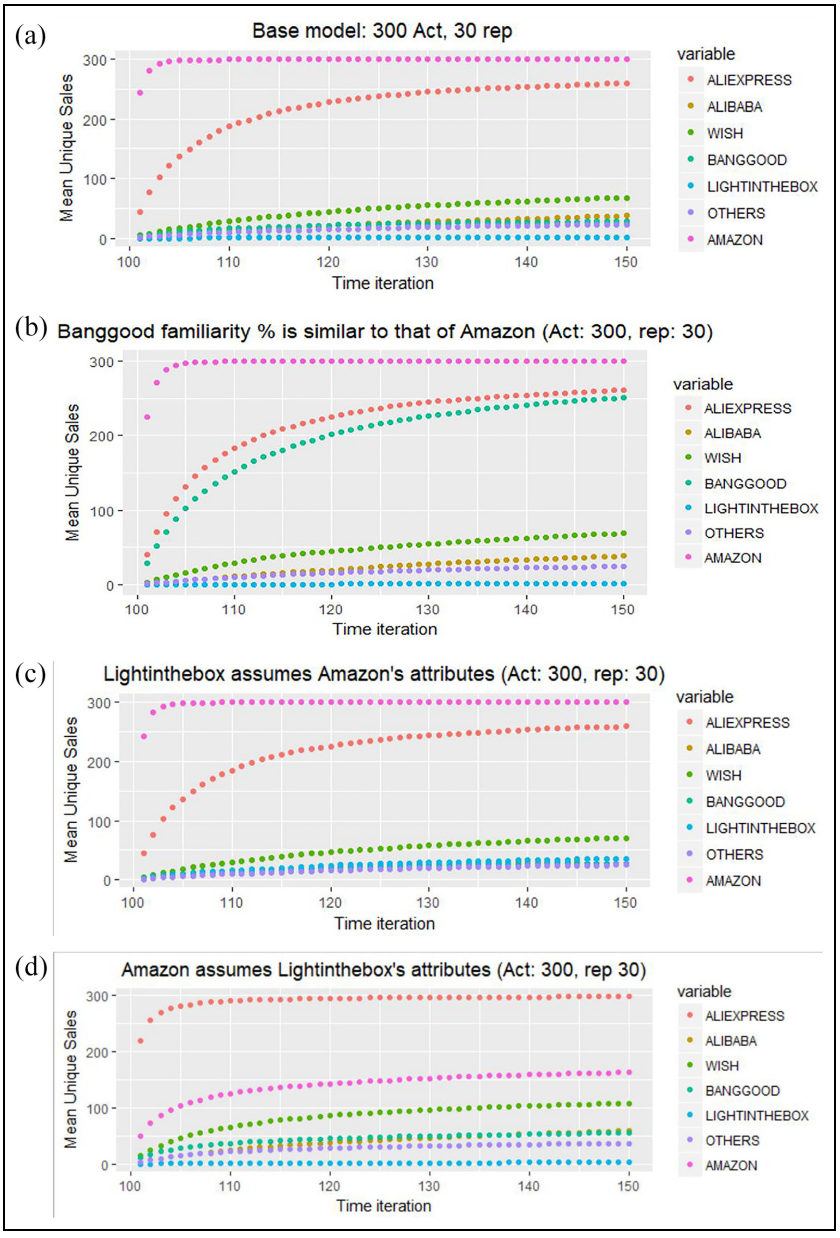

Figure 6. Scenarios for observing the effect of the percentage of buyers familiar with a marketplace and marketplace attributes in the structure of the market by observing *unique sales*. The graphs present the following: (a) the model base; (b) Banggood is as well-known as Amazon; (c) LightInTheBox has Amazon's attributes; and (d) Amazon has LightInTheBox's attributes. Like in previous experiments, data were collected at over 50 iterations in all cases, from iteration 101 to iteration 150.

In summary, on one hand, increasing ''how well-known Banggood is'' significantly improves its performance; on the other hand, a marketplace that is as well-known as LightInTheBox and with the (positive) attributes of Amazon (Graph b) does not significantly improve its performance in terms of the variable *unique sales* or market share, while a marketplace that is as well-known as Amazon but with the attributes of LightInTheBox significantly improves its performance, helping the firm reach a much better position in the market than in the previous scenario. Therefore, ''how well-known a marketplace is'' is a much stronger factor and changes the performance of the marketplace more than an improvement of its attributes. Consequently, in order of importance, the factors that drive market share in cross-border B2C e-commerce are (1) how well-known the company is, (2) marketplace attributes, and (3) WOM.

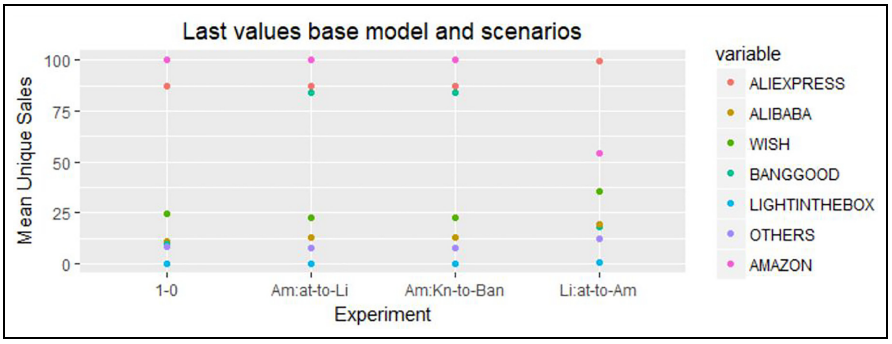

Figure 7. Final values in the simulation for the scenarios generated to observe the effect of the percentage of buyers familiar with a marketplace and attributes of the marketplace in the structure of the market for several scenarios. At the horizontal edge: Am: at-to-Li: Amazon's attributes are given to LightInTheBox; Am: kn-to-Ban: the "how well-known Amazon is" value is assigned to Banggood; Li: at-to-Am: LightInTheBox's attributes are given to Amazon. Figure 6's graphs show the value of the mean of unique sales for each marketplace and scenario at a simulation iteration of 150.

4.4. Comparing our method with other techniques

Our method for proving tendencies (oligopoly, in this paper) combines theorem-proving, scenario analysis, and domain-expert knowledge. This method shares similarities with the constrained exploration of simulation trajectories approach described in the section "Constraint proof of tendencies in simulation outputs," but it also incorporates domain expertise and is applied to a real-world case (unlike the ideal model used in the method referenced in that section). In contrast to the method used in Leger's study,[14] which was restricted to Chinese B2C marketplaces, our approach studies B2C marketplaces globally. In addition, the sections "A practical proof" and "Relative importance of the factors driving the tendency" outline procedures to verify the generality of the tendency, demonstrating the robustness of the conclusions and results and differentiating the significance of the factors driving the tendency.

Our approach addresses some of the challenges inherent in methods for proving tendencies in simulation (see the section "Proving tendencies in a model")[8,9,54]:

(1) It effectively manages the complexity of the proof by constraining the exploration of simulation trajectories to the parameter subsets suggested by domain experts and the statistically valid random exploration of agent choices, thus exploring all potential states;
(2) It is expressive, using domain-specific language while remaining decidable, providing flexibility in defining relevant tendencies and ensuring the study's pertinence;
(3) The model represents a real system with minimal abstraction, making it simple to analyze and validate;
(4) The proof is scalable, controlling the demand for processing power and memory by preventing the explosion of the number of simulation branches explored;
(5) Our procedure can be integrated with or developed alongside other simulation tools;
(6) Our method does not require extensive expertise in formal methods, including logic and mathematics;
(7) As demonstrated by the case study, our method can be applied to real scenarios while maintaining a high level of accuracy;
(8) The low complexity of our method makes it a good candidate for application in evolving systems.

Among the limitations of our method is that the exploration of simulation trajectories is random, using scenario analysis, and domain-expert-oriented rather than relying on logical or mathematical formality. Thus, our method is less formal than most theorem-provers and could be seen as a weaker proof than others in the literature. This weakness is offset by its advantages, which stem from combining simulation-branch exploration with methods like domain expert advice, scenario analysis with varying numbers of simulation replications, and variations in the number of agents.

5. Conclusion and future work

The paper has achieved its two primary goals: developing a method for proving tendencies in a simulation and demonstrating the oligopolistic nature of cross-border B2C e-commerce marketplaces.

First, the paper introduced a method for identifying and proving a tendency, showcasing the robustness of the conclusions and simulation model, and discerning the significance of the factors driving the tendency. This method integrates theorem-proving, scenario analysis, and domain-expert knowledge as follows. The explored tendency was identified by examining simulation trajectories based on

scenario analysis and the Monte Carlo method. The scenarios were defined by domain experts (the study's researchers), while agent choices were probabilistically determined in simulation-trajectory replications for each parameter configuration. We demonstrated that the tendency persists if (1) the number of replications increases and (2) the number of simulated agents is augmented. The persistence of the tendency with more agent choices indicates its necessity, while its consistency with more agents validates the model regardless of the number of simulated agents. A second set of scenarios was later analyzed to determine the relative importance of factors driving market share. This method is more practical than others, as it requires exploring fewer simulation trajectories than, for example, an exhaustive constraint proof of tendencies in the simulation outputs. This practicality was illustrated by comparing the number of required simulation trajectories for each method in several cases.

Our approach addresses the challenges inherent in proving tendencies in simulation by effectively managing the complexity of the proof through the constrained exploration of simulation trajectories to the parameter subsets suggested by domain experts and the statistically valid random exploration of agent choices. In addition, it is expressive, flexible, scalable, easily integrated with other simulation tools, simple (not requiring extensive expertise in formal methods), and applicable to real cases and evolving systems.

Second, this method allowed us to demonstrate the oligopolistic character of cross-border B2C e-commerce marketplaces—a complex system—and the order of importance of the factors driving market share: (1) marketplace recognition, (2) product and service attributes, and (3) WOM. In addition, the model was validated against survey data.

Oligopoly is a market structure in which a small number of firms dominate and control the majority of market share. Their strategies can affect both their own market shares and that of their competitors. We have shown that Amazon and AliExpress are strong market leaders, with competitors trailing far behind. This tendency indicates that the most well-known marketplaces will initially maintain a larger market share during the simulation process. During the experimental process, interesting partial results were observed; for example, Amazon's performance remained strong even with the worst attributes of other marketplaces in the model, dropping to second place, behind AliExpress, only in the worst-case scenario.

This conclusion aligns with data from previous studies, indicating that the cross-border B2C market is oligopolistic, with a few marketplaces holding over 50% of the market share. Amazon is the global leader, followed by Chinese platforms like AliExpress. Changes in strategy by one marketplace will affect its market share and those of its competitors, making strategic suggestions from scenario analysis relevant.

Amazon, AliExpress, and to some extent Wish (in third place) benefit from being the most well-known marketplaces. This reality often leads consumers to overlook other marketplaces. In addition, Amazon and AliExpress have positive attributes, making it challenging for competitors to reach their level. Our results suggest that the most effective strategy for competitors to approach these market leaders is to increase their recognition among consumers. This strategy would be particularly beneficial for Banggood, a lesser-known marketplace with good attributes, as demonstrated in several scenarios.

This paper provides, on one hand, a specific theorem-proving method useful for the social-simulation community dealing with real-world cases and, on the other hand, an important contribution to those disciplines interested in markets, particularly B2C marketplaces, by presenting evidence that B2C marketplaces are oligopolies.

One limitation of our theorem-proving method is that most of its exploration of simulation trajectories is random (using the Monte Carlo method) and domain-expert-knowledge-oriented. This results in a less-formal proof than most of the usual theorem-provers found in the literature, and it may be considered a weak proof by some researchers working with formal methods. Future work could include developing facilities for automatically proving tendencies (conducting scenario analysis and Monte Carlo exploration of agent choices) on an open-access website. These facilities would complement the simulation tool we have already developed.

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Appendix 1

Table 7. Percentage of buyers familiar with each marketplace.

Marketplace	Proportion of buyers familiar with the marketplace
Amazon	100,00
AliExpress	92.27
Banggood	10.14
Wish	41.30
Others	14.25
LightInTheBox	11.59
Alibaba	42.51

Table 8. Importance of the attributes for buyers. These are the average importance values of the attributes given by the surveyed participants. From these values, the endorsements for each attribute are obtained (see Entities, state variables, and scales in section 3.2).

Attribute	Importance of attribute for buyers
Reputation of the marketplace	5.94
Company size	4.77
Website appearance	5.29
Website content quality	6.05
Website technical quality	5.74
Security of the web	5.65
Privacy	4.78
Consumer reviews	6.06
Cost saving	6.19
Product offering	5.99
Product quality	6.25
Service quality	6.08
Previous experience with the website (satisfaction)	5.99

Table 9. Low (L) and high (H) values of the attributes shown by the marketplaces, as perceived by buyers. Thus, the frequency for low was obtained by counting the number of answers in the survey in the interval [1, 4] (half of the whole interval: (1, 7) The probability of a low valuation was calculated by dividing this frequency by the total number of answers. The probability for a high valuation is the complement of the probability for a low one (see section 3.2).

Attribute/ zmarketplace	AliExpress		Alibaba		Wish		Banggood		LightInTheBox		Others		Amazon	
	L	H	L	H	L	H	L	H	L	H	L	H	L	H
Reputation of the B2C marketplace	0.338	0.662	0.528	0.472	0.55	0.45	0.452	0.548	0.583	0.417	0.542	0.458	0.083	0.917
Company size	0.134	0.866	0.205	0.795	0.269	0.731	0.381	0.619	0.271	0.729	0.407	0.593	0.054	0.946
Website appearance	0.285	0.715	0.472	0.528	0.409	0.591	0.381	0.619	0.563	0.437	0.441	0.559	0.171	0.829
Website content quality	0.393	0.607	0.591	0.409	0.591	0.409	0.452	0.548	0.583	0.417	0.525	0.475	0.086	0.914
Website technical quality	0.246	0.754	0.432	0.568	0.409	0.591	0.238	0.762	0.417	0.583	0.356	0.644	0.064	0.936
Security of the web	0.298	0.702	0.466	0.534	0.474	0.526	0.286	0.714	0.521	0.479	0.373	0.627	0.076	0.924
Privacy	0.401	0.599	0.54	0.46	0.213	0.787	0.5	0.5	0.604	0.396	0.576	0.424	0.257	0.743
Consumer reviews	0	1	1	0	0	1	0	1	0	1	0	1	0	1
Cost saving	0.217	0.783	0.165	0.835	0.123	0.877	0.238	0.762	0.125	0.875	0.186	0.814	0.482	0.518
Product offering	0.131	0.869	0.273	0.727	0.211	0.789	0.19	0.81	0.271	0.729	0.254	0.746	0.071	0.929
Product quality	0.531	0.469	0.676	0.324	0.719	0.281	0.524	0.476	0.708	0.292	0.627	0.373	0.122	0.878
Service quality	0.67	0.33	0.653	0.347	0.749	0.251	0.619	0.381	0.792	0.208	0.797	0.203	0.222	0.778

Table 10. Parameters defining the overall configuration of the model.

Variable or factor	Description	Value in scenario 0
CONTACTS	Mean number of individuals a person communicates with through WOM	17
LEARNING_PERIODS	Number of time iterations used for the unstable initial period before the studied stable period	100
PERIODS	Total number of simulated iterations (100 for the unstable period and 200 for the stable period studied)	300
BASE	Base used for the buyers' learning function (endorsements function)	1.2
FRIENDS	Proportion of contact with individuals an agent communicates with (WOM)	0.7
MEMORY	–1 means that the agents never forget an endorsement. In cases when a different value is used, it will indicate the number of iterations an endorsement is remembered	–1
WOM	0 if WOM is not activated, 1 otherwise.	1
AGENTS	Number of buyers	50
GUI		1
COMPRESSED_RESULTS		1
MARKET_QUOTA		1
SAVED_AGENT_DECISIONS		0
REPETITIONS	Number of simulation replications	29

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